

COUNTING HALLUCINATIONS IN DIFFUSION MODELS

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ABSTRACT

Diffusion probabilistic models (DPMs) provide strong generative performance for modalities such as images and videos. Nevertheless, their samples can contain hallucinated content (*hallucinations*) that is inconsistent with factual or physical constraints, for example an implausible duplicate cup floating adjacent to another cup. Although such failures are frequently observed, the absence of tractable procedures for measuring them systematically limits empirical analysis and constrains the development of generative models with explicit factual validity requirements. We study a concrete and measurable subclass of this phenomenon, which we denote **counting hallucination**: the generation of an invalid number of object instances or structured components, such as a hand image with six fingers, even when the corresponding count pattern is absent from the training distribution. We introduce **CountHalluSet**, a dataset suite with explicit count constraints, consisting of ToyShape, SimObject, and RealHand. Using this suite, we define a standardized evaluation protocol for estimating counting hallucination rates and evaluate how DPM sampling conditions—including solver family, ODE solver order, number of sampling steps, and initial noise realization—affect the measured rate. We further quantify the association between counting hallucinations and standard image-generation metrics, including Fréchet Inception Distance (FID), and show that widely used perceptual metrics do not consistently represent counting-specific hallucination severity.

1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), commonly referred to as diffusion models, constitute a dominant class of generative models and produce high-quality samples across text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). Despite this empirical progress, diffusion models do not always preserve factual attributes implicit in the training distribution; generated outputs can violate real-world consistency constraints. One representative failure is the synthesis of entities with an invalid number of spatially structured parts, including extra human fingers (Narasimhaswamy et al., 2024) or additional animal limbs (Borji, 2023).

Prior work has characterized a range of failure modes in text-guided diffusion systems. For example, Borji (2023) provide a systematic qualitative taxonomy of deficiencies in text-to-image models. Complementary studies introduce procedures for discovering undesirable behaviors: Liu et al. (2024) formulate adversarial search for failure-case discovery, whereas Wu et al. (2023) use an iterative procedure to identify and mitigate spurious correlations.

The hallucination failure mode in diffusion models has recently received more explicit attention (Aithal et al., 2024). In contrast to prompt-conditioned errors, which can be evaluated through input-output alignment metrics such as CLIP score (Hessel et al., 2021) or VQA score (Huang et al., 2025; Zarei et al., 2024), and in contrast to visually obvious defects such as severe distortion, blur, or artifacts, hallucinated samples (*hallucinations*) may remain perceptually plausible while violating factual constraints encoded by the training data. Examples include incorrect object multiplicities, visually plausible but physically infeasible geometries, and other violations of physical law. The distinction between a valid sample and a hallucinated one is therefore fine-grained and operationally difficult for current AI systems to determine reliably. For instance, even state-of-the-art large vision-

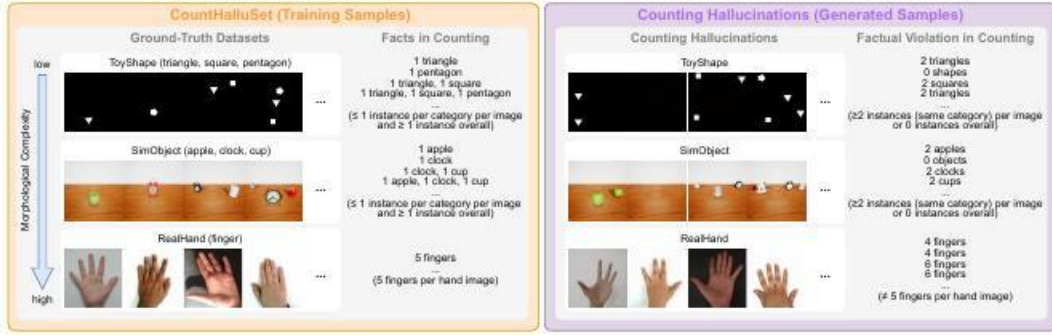


Figure 1: Comparison between reference samples from the proposed **CountHalluSet** training suite (left) and corresponding model-generated **counting hallucinations** (right). Each dataset imposes explicit count criteria over its object categories. In ToyShape and SimObject, each image contains at most one instance per category and at least one instance in total; in RealHand, each image contains exactly five fingers. Counting hallucinations are generated samples that violate these dataset-specific count constraints, such as an image with two apples or an image with no objects on an otherwise normal background, although these configurations are absent from the SimObject training data.

language models such as GPT-4o can fail on the elementary perceptual operation of counting fingers in an otherwise clear hand image. A standardized protocol with explicit validity criteria is therefore required for reliable measurement of hallucinated samples, and such measurement is a prerequisite for designing next-generation generative models with stronger factual consistency, an ability central to their use as world models (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

Hallucinations can involve counts, colors, spatial relations, object geometry, or physical realism. Among these categories, counting errors are especially amenable to controlled analysis because the relevant property is discrete, directly inspectable, and quantitatively measurable. We therefore formalize a restricted hallucination type, termed **counting hallucination**. *In this setting, the admissible object counts in the training data are specified a priori. A generated image is classified as hallucinated whenever its object counts fall outside these prescribed values.* For example, a generated hand with six fingers is a counting hallucination relative to RealHand because no such configuration occurs in that dataset, as illustrated in the final row of Fig. 1. To support controlled evaluation, we construct CountHalluSet, a dataset suite comprising three datasets whose countable objects vary in morphological complexity: ToyShape (triangle, square, pentagon), SimObject (mug, apple, clock), and RealHand (finger).

For each dataset, we define explicit counting criteria and evaluate generated samples against those criteria. ToyShape and SimObject are synthetic datasets in which each category may appear at most once and each image must contain at least one object. RealHand is a real-image dataset in which each valid sample contains exactly five fingers. These datasets allow us to measure counting hallucinations under increasing visual and structural complexity while keeping the admissible count space well specified.

Figure 1 illustrates representative examples from these datasets alongside corresponding counting hallucinations.

This paper makes the following contributions. First, we introduce the notion of counting hallucination as a formally specified and empirically measurable subclass of diffusion-model hallucinations. Second, we construct CountHalluSet and provide dataset-specific counting criteria, counting models, and evaluation protocols for quantifying counting hallucination rate (CHR). Third, we conduct controlled experiments over solver type, solver order, sampling steps, and initial noise to characterize how denoising conditions affect CHR. Fourth, we analyze correlations between CHR, non-counting failure rate (NCFR), total failure rate (TFR), and standard perceptual metrics, showing that metrics such as FID Heusel et al. (2017) and MUSIQ Ke et al. (2021) do not consistently capture count-specific factual failures. Finally, we evaluate pretrained text-to-image hand generation and a joint diffusion training strategy, demonstrating both the broader relevance of the phenomenon and a practical direction for reducing counting hallucinations.

2 RELATED WORK

2.1 HALLUCINATIONS IN DIFFUSION MODELS

Motivated by the extensive literature on hallucination in large language models (LLMs) (Kadavath et al., 2022; Ji et al., 2023; Huang et al., 2023), recent work has begun to examine analogous hallucination phenomena in unconditional diffusion models. In particular, Aithal et al. (2024) propose a qualitative hypothesis in which hallucinated samples arise from data configurations interpolating between distinct modes. This hypothesis provides a plausible mechanistic account, but it does not by itself supply a systematic quantitative protocol for measuring hallucination frequency or severity in diffusion-generated samples.

The present study addresses this methodological gap by specifying count-based validity constraints and measuring their violation in generated outputs. Such an analysis supports a more precise characterization of undesirable diffusion-model behavior and permits controlled evaluation of how denoising-related factors, including the number of sampling steps and the order of ODE solvers, modulate hallucination rates.

2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

VQA score (Huang et al., 2025; Zarei et al., 2024) and CLIP score (Hessel et al., 2021) are commonly used to evaluate text-guided image generation through language-conditioned input-output alignment. However, metrics that depend on language-based models are structurally limited when the objective is to detect hallucinated visual content. One reason is that pretrained language models exhibit a nonzero lower bound on hallucination propensity for certain fact types, which in turn constrains the attainable accuracy of hallucination detection systems based solely on such models (Kalai & Vempala, 2024). A second reason is the persistent gap between human annotators and LLM-based systems when classifying hallucinated textual content (Lin et al., 2022; Li et al., 2023).

Image generation is also commonly evaluated with distributional or perceptual metrics such as FID (Heusel et al., 2017), IS (Salimans et al., 2016), Precision & Recall (Sajjadi et al., 2018), and MUSIQ (Ke et al., 2021). These metrics typically operate on feature embeddings extracted by a pretrained Inception-v3 model (Szegedy et al., 2016) or on learned perceptual-quality predictors. Whether such metrics faithfully represent the severity of hallucinations in generated datasets remains unresolved. We investigate this question for the specific case of counting hallucinations, where the validity condition is explicitly defined by dataset-level count constraints.

3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

3.1 FORWARD (DIFFUSION) PROCESS

Forward Process. Given observed data $\mathbf{x}_0 \sim q_0(\mathbf{x}_0)$, Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) define a *forward process* (also called the *diffusion process*) that progressively corrupts data by injecting Gaussian noise over T discrete timesteps. At timestep t , the marginal conditional distribution of the noisy variable $\mathbf{x}_t$ given the clean sample $\mathbf{x}_0$ is

$$q_{0:t}(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t | \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

where $t \in \{0, 1, \dots, T\}$ and $\{\beta_i\}_{i=1}^T$ is a predefined timestep-dependent noise schedule, typically instantiated as a linear (Ho et al., 2020) or cosine (Nichol & Dhariwal, 2021) schedule. The scalar $\bar{\alpha}_t$ determines the signal-to-noise ratio at step t : $\bar{\alpha}_0 = 1$ corresponds to the clean-data limit, whereas $\bar{\alpha}_T \approx 0$ corresponds to an approximately pure-noise distribution. Equivalently, Eq. (1) can be represented by the following reparameterization:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

For sufficiently large T , the marginal distribution $q_{0:T}(\mathbf{x}_T | \mathbf{x}_0)$ approaches the standard Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. In commonly used DPMs (Ho et al., 2020; Nichol & Dhariwal, 2021), T is set to 1000.

3.2 REVERSE (DENOISING) PROCESS.

Training Objective. The *reverse process* (also called the *denoising process*) aims to reconstruct samples from the data distribution by iteratively transforming the terminal noisy variable $\mathbf{x}_T$ toward $\mathbf{x}_0$. DPMs parameterize this procedure by training a neural network $\epsilon_\theta(\mathbf{x}_t, t)$, referred to as the *noise prediction model*, to estimate the injected noise ϵ from $\mathbf{x}_t$ and t . The parameter vector θ is optimized by minimizing the weighted denoising objective (Ho et al., 2020; Song et al., 2022):

$$\min_{\theta} \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\pi(t) \|\epsilon_\theta(\mathbf{x}_t, t) - \epsilon\|_2^2], \quad (3)$$

where $\pi(t) > 0$ denotes a timestep weighting function.

Ancestral Sampling. Given terminal noisy data $\mathbf{x}_T$, samples from the original distribution $q_0(\mathbf{x}_0)$ can be generated by reversing the diffusion chain with the same number of discrete steps; this procedure is typically called *ancestral sampling*:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \frac{1}{\sqrt{1-\beta_t}}(\mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)), \beta_t \mathbf{I}), \quad (4)$$

where $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ is the only unknown quantity and is referred to as the *score function* (Song et al., 2022).

In practice, DPMs approximate the scaled score function $-\sqrt{(1-\bar{\alpha}_t)} \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ using the learned noise prediction model. This yields the following equivalent sampling update associated with Eq. (3):

$$\mathbf{x}_{t-1} = \mu_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon, \quad \mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1-\beta_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t). \quad (5)$$

ODE-Based Accelerated Sampling. The ancestral update in Eq. (4) contains a stochastic term (Kloeden et al., 1992). It also admits an associated deterministic process expressible as an ordinary differential equation (ODE), commonly referred to as the *probability flow ODE* (Song et al., 2022):

$$\frac{d\mathbf{x}(\tau)}{d\tau} = f(\tau)\mathbf{x}(\tau) + h(\mathbf{x}(\tau), \tau), \quad f(\tau) = -\frac{1}{2}\beta(\tau), \quad h(\mathbf{x}(\tau), \tau) = -\frac{1}{2}\beta(\tau) \nabla_{\mathbf{x}(\tau)} \log q_\tau(\mathbf{x}(\tau)), \quad (6)$$

where $\tau \in [T, 0]$. Let $\mathbf{x}(\tau)$ denote the ideal trajectory that satisfies this ODE under the true score term $h(\mathbf{x}(\tau), \tau)$ and is initialized at an ideal diffused sample $\mathbf{x}(T)$ drawn from the true terminal distribution $q_T(\mathbf{x}(T))$. Define the discrete grid by $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, with $\tau_N = 0$. Applying the variation-of-constants formula gives the exact one-step solution

$$\mathbf{x}(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h(\mathbf{x}(u), u) du \quad (7)$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state-transition operator for the linear subsystem $\frac{dz}{d\tau} = f(\tau)z$. The step size $\Delta t > 0$ permits ODE-based samplers to use larger discretization intervals than ancestral sampling, yielding a total of $T/\Delta t$ sampling steps. Because the linear component $\mathcal{G}(t_2, t_1)$ is available in closed form, accelerated ODE-based samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022; 2023) primarily allocate numerical approximation effort—for example through higher solver order or additional steps—to the nonlinear integral involving h . Additional theoretical details are provided in Appendix A.1 to A.3.

4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

This section specifies the objects required to measure counting hallucinations: dataset-level counting criteria, the induced definition of a counting hallucination, and the evaluation protocol used for each dataset in CountHalluSet. We first formulate the validity constraints that define admissible object counts, then instantiate those constraints for ToyShape, SimObject, and RealHand. For each dataset, we identify the corresponding violation set and describe the counting model or filtering procedure used to estimate hallucination rates.

4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

We define counting hallucinations as generated samples whose measurable object counts violate count constraints specified by the reference training dataset. This formulation treats hallucination detection as a constrained counting problem rather than as a generic perceptual-quality judgment.

Counting criteria in a training dataset. Let $\mathcal{D}_{\text{ref}}$ denote a reference training dataset. For any reference sample $\mathbf{x} \in \mathcal{D}_{\text{ref}}$, let $N_c(\mathbf{x})$ be the number of objects of category c in $\mathbf{x}$. The admissible values of $N_c(\mathbf{x})$ are specified by a predefined set $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$ (e.g., $\{0, 1\}$ or $\{1, 2, 3\}$), such that $N_c(\mathbf{x}) \in \mathcal{S}_c$. We additionally impose a nonempty-sample constraint, $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$, where C denotes the set of evaluated categories.

Counting hallucinations as deviations from the counting criteria. For a generated sample $\hat{\mathbf{x}}$, we say that $\hat{\mathbf{x}}$ exhibits a *counting hallucination* with respect to $\mathcal{D}_{\text{ref}}$ when the following two conditions are simultaneously satisfied:

1. **Sufficient visual quality for counting.** The generated image $\hat{\mathbf{x}}$ must have sufficient visual quality for the background and any present objects to be unambiguously discerned. This condition separates count violations from other failure modes, such as severe degradation or uninterpretable structure. We denote this condition by a binary indicator $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \in \{0, 1\}$, called the *counting-ready indicator*; value 1 indicates that the image is suitable for counting, whereas value 0 indicates that it is not. In implementation, this indicator may be assigned by human annotation, by a classifier calibrated to human judgments, or by another automated criterion.
2. **Violation of counting facts.** A count violation occurs if there exists a category c such that $N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$, or if the sample contains no objects, i.e., $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$, despite the reference dataset requiring at least one object per sample. We operationalize this condition by applying a counting model to generated images and using the predicted counts as evidence of violation.

A counting hallucination (CH) is therefore detected when the sample is countable ($\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) = 1$) and its predicted counts violate the dataset-specific criteria ($\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$ or $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$). Formally,

$$\mathbb{I}_{\text{CH}}(\hat{\mathbf{x}}) = \mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \wedge \left(\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c \vee \sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0 \right).$$

4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

CountHalluSet contains three datasets designed for controlled measurement of counting hallucinations: ToyShape, SimObject, and RealHand. Dataset examples and corresponding hallucination examples are shown in Fig. A1 and A2, respectively. Additional dataset statistics are reported in Appendix A.4.

4.2.1 TOYSHAPE

Dataset description. ToyShape consists of 30,000 images containing three geometric categories: *triangle*, *square*, and *pentagon*. All shapes are white, have equal area of 120 pixels, and are arranged without overlap. The counting rule requires that each image contain at most one instance of each shape category and at least one shape in total, as shown in the upper-left panel of Fig. 1. Formally, $C = \{\text{triangle}, \text{square}, \text{pentagon}\}$. For each class $c \in C$, we set $N_c(\mathbf{x}) \in \mathcal{S}_c = \{0, 1\}$ subject to $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Counting hallucinations. Under these criteria, any sample generated by a model trained on ToyShape that contains two or more instances of the same shape category is a counting hallucination; an example is the image with two pentagons in the upper-right region of Fig. 1. Empty images containing only the black background and no shapes also violate the nonempty-sample constraint and are therefore counted as hallucinations. Additional examples are provided in Appendix A.5.

Counting model used in ToyShape. Because ToyShape is visually simple and severe degradations are rare, we evaluate all generated images directly without a separate counting-ready indicator. We

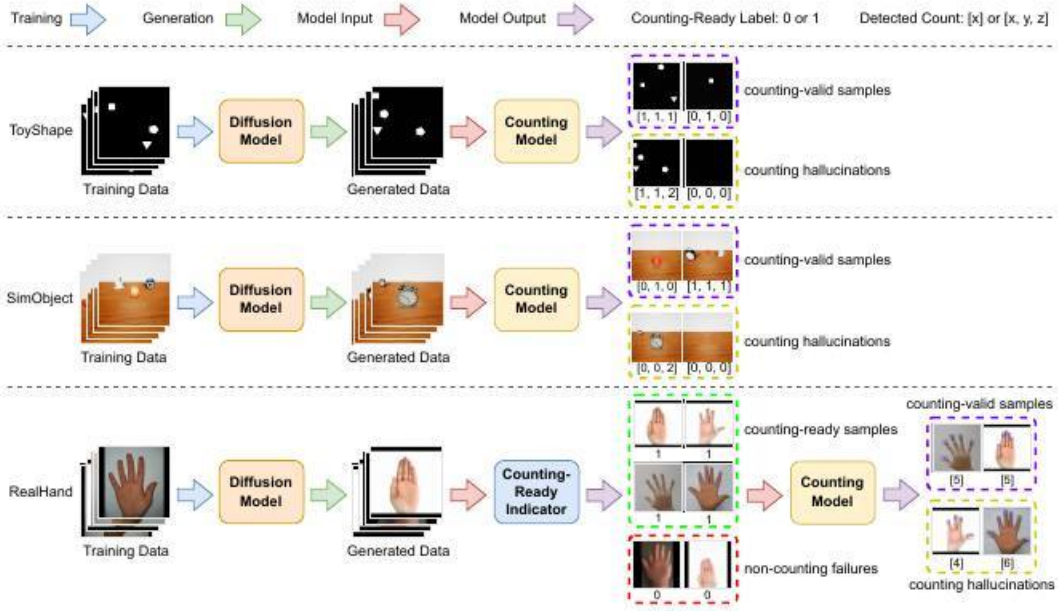


Figure 2: Evaluation protocol for the CountHalluSet suite. For ToyShape and SimObject, generated images are submitted directly to the corresponding counting model. For RealHand, a counting-ready indicator, implemented as a binary classifier aligned with human judgments, first separates countable hand images from non-counting failures such as severely deformed fingers. Count labels are ToyShape: *triangle*, *square*, *pentagon*; SimObject: *mug*, *apple*, *clock*; and RealHand: *finger*.

fine-tune a ResNet-50 (He et al., 2016) counting model using more than 400k ToyShape samples in which each category appears 0–3 times per image (i.e., $S_c = \{0, 1, 2, 3\}$). This expanded count support exposes the model to both valid samples and plausible count violations, enabling accurate quantification of generated outputs. The resulting model achieves more than 99.9% counting accuracy on generated data.

Quantification of counting hallucinations. For a diffusion model trained on ToyShape, we generate 30,000 samples, matching the size of the training dataset. The counting model predicts category-wise object counts for each generated image. A generated image is then labeled as a counting hallucination exactly when the predicted counts violate the ToyShape criteria, as illustrated in Fig. 2.

4.2.2 SIMOBJECT

Dataset description. SimObject contains 30,000 rendered images of everyday objects from three categories: *mug*, *apple*, and *clock*. Each category includes 10 intra-class variations. Objects are randomly placed on a wooden tabletop under fixed illumination, with little or no mutual occlusion, as shown in the middle-left panel of Fig. 1. The dataset is synthesized in Unreal Engine 5 (Epic Games, 2024), using photorealistic rendering with explicit simulation of object geometry, material properties, lighting, and reflections. Its counting criteria match the ToyShape criteria (see Sec. 4.2.1). Formally, $C = \{mug, apple, clock\}$. For each class $c \in C$, we set $N_c(\mathbf{x}) \in S_c = \{0, 1\}$ subject to $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Quantification of counting hallucinations. Because SimObject shares the same count constraints as ToyShape, the hallucination definition, counting-model construction, and quantification protocol follow the ToyShape procedure, as summarized in Fig. 2.

4.2.3 REALHAND

Dataset description. To evaluate counting hallucinations under real-image conditions, we construct RealHand with 5,050 human hand images sourced from the 11k Hands dataset (Afifi, 2019), Kaggle¹,

¹<https://www.kaggle.com/>

and Roboflow². We aggregate raw images from these sources and remove images exhibiting blur or visible artifacts. Each retained image depicts one human hand with five fingers and includes dorsal (back) and palmar (front) views under varied lighting and background conditions, including white surfaces, blackboards, and walls, as shown in the lower-left panel of Fig. 1. The counting criterion requires exactly five fingers per image: $C = \{finger\}$ with $N_c(\mathbf{x}) \in S_c = 5$ for each $c \in C$, and $\sum_{c \in C} N_c(\mathbf{x}) = 5$.

Counting hallucinations and non-counting failures. Realistic hand generation produces failures beyond count violations, including blur, severe finger deformation, and other visual artifacts, in addition to extra or missing fingers. We therefore distinguish countable samples from non-counting failures using a counting-ready indicator implemented as a binary classifier. Examples of counting hallucinations and non-counting failure samples are provided in Appendix A.7.

Counting-ready indicator and counting model used in RealHand. To separate counting hallucinations from the preceding non-counting failure modes, we fine-tune a MaxViT (Tu et al., 2022) model on 2.5k annotated generated images and use it as a counting-ready indicator aligned with human judgments. For finger counting, we fine-tune a YOLO-12 (Tian et al., 2025) detector on 2k generated images with fingertip annotations. On a human-evaluated set of 500 randomly sampled images annotated by three annotators, the indicator and counting model achieve more than 96% and 99% accuracy, respectively. YOLO detection examples are shown in Appendix A.6.

Quantification of counting hallucinations. Unlike ToyShape and SimObject, where all generated samples can be evaluated directly because severe non-counting failures are rare, RealHand requires a preliminary filtering step. We first apply the counting-ready indicator to select hand images with sufficient visual quality for counting; the counting model is then applied only to these selected images to predict the number of fingers, as shown in Figure 2.

5 EXPERIMENTS

5.1 IMPLEMENTATION DETAILS

Training of Diffusion Models. For ToyShape, we train a DDPM (Ho et al., 2020) from scratch. For SimObject and RealHand, we fine-tune latent diffusion models (LDMs) (Rombach et al., 2022) initialized from CelebA-HQ-pretrained weights (Karras et al., 2018). All models use $T = 1,000$ diffusion steps with the standard linear noise schedule. Input images are resized to 128×128 for DDPM training and to 256×256 for LDM training. Optimization uses Adam (Kingma, 2015) with learning rate 0.0001, effective batch size 256, and training lengths of 150k, 300k, and 80k steps for ToyShape, SimObject, and RealHand, respectively.

Inference of Diffusion Models. For ancestral sampling (DDPM), we use the full 1,000-step reverse chain. For ODE-based sampling, we evaluate a first-order method, DPM-Solver-1 (equivalent to DDIM (Song et al., 2021)), and a second-order method, DPM-Solver-2 (Lu et al., 2022). Each ODE solver is evaluated with 25, 50, and 100 denoising steps. For RealHand, the YOLO counting model uses confidence threshold 0.3 and IoU threshold 0.1. Unless otherwise specified, each evaluation generates the same number of samples as the corresponding training set, and reported values are averaged over three random seeds.

5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

Table 1 reports counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) across solver families, ODE solver orders, sampling-step budgets, and initial-noise settings for ToyShape, SimObject, and RealHand. The results support the following observations.

(1) **Additional sampling steps reduce CHR for synthetic datasets but can increase CHR for RealHand.** In Table 1, increasing the ODE sampling budget from 25 to 50 to 100 steps generally decreases CHR on ToyShape and SimObject, whereas the same change tends to increase CHR on RealHand, except for the 100-step DPM-Solver-2 configuration. This discrepancy indicates that simpler synthetic distributions benefit from finer numerical discretization, while the more complex

²<https://universe.roboflow.com/>

Table 1: Counting hallucination rate (CHR; counting hallucinations divided by total generated samples) under solver and initial-noise configurations for ToyShape, SimObject, and RealHand. For RealHand, non-counting failure rate (NCFR) denotes failures rejected by the counting-ready indicator. Total failure rate (TFR) is computed as CHR plus NCFR. *Diffused* denotes ground-truth initial noise, whereas *Normal* denotes standard Gaussian initial noise.

Solver Name	Sampling Steps	CHR (ToyShape)		CHR (SimObject)		CHR (RealHand)		NCFR (RealHand)		TFR (RealHand)	
		<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>
DPM-Solver-1 (Song et al., 2021)	25	2.43	3.47	9.27	9.95	12.71	12.95	16.18	18.06	28.90	31.01
	50	1.83	2.79	8.53	9.27	13.82	13.85	11.32	12.51	25.14	26.36
	100	1.56	2.32	8.04	8.90	14.29	14.55	9.54	10.63	23.83	25.19
DPM-Solver-2 (Lu et al., 2022)	25	2.45	3.81	8.16	8.19	14.23	14.48	8.37	9.33	22.60	23.82
	50	1.76	2.86	7.90	7.95	16.15	15.99	6.38	7.22	22.53	23.21
	100	1.41	2.24	7.89	7.89	15.06	15.43	7.82	8.94	22.88	24.38
DDPM (Ho et al., 2020)	1000	0.63	0.64	4.98	4.99	10.82	10.75	2.07	2.39	12.88	13.14

RealHand distribution may expose local inconsistencies that additional denoising steps amplify into count violations.

(2) **Higher-order ODE solvers reduce NCFR but increase CHR for RealHand under standard step budgets.** As shown in Table 1, the RealHand comparison between DPM-Solver-1 and DPM-Solver-2 at 25 and 50 steps exhibits a systematic trade-off: the second-order solver substantially decreases non-counting failures but yields a higher counting hallucination rate. A related pattern appears when the sampling-step budget is increased for either DPM-Solver-1 or DPM-Solver-2. Thus, increased numerical refinement improves overall visual interpretability, as reflected by lower NCFR, but does not enforce the factual constraint that a valid hand image contain exactly five fingers.

(3) **DDPM yields the lowest observed CHR, NCFR, and TFR.** Across the evaluated settings, ancestral sampling (DDPM) consistently attains the smallest CHR, NCFR, and TFR. These measurements indicate that the full stochastic reverse chain provides the strongest empirical baseline among the considered samplers for reducing both counting hallucinations and non-counting failures.

(4) **Initial-noise alignment reduces both counting and non-counting failures.** As reported in Table 1, “diffused” noise, i.e., ground-truth initial noise, generally produces lower CHR, NCFR, and TFR than “normal” standard Gaussian noise. This effect is consistent with the interpretation that diffused noise better matches the training-time terminal distribution, so a more distributionally aligned initialization can reduce both count violations and other visual failures.

(5) **Object morphology is associated with increased counting-hallucination risk.** The cross-dataset comparison from ToyShape to SimObject to RealHand shows higher CHR as the target objects become more morphologically complex. This pattern suggests that fine-grained local structure and complex spatial relationships make it harder for diffusion models to preserve discrete object counts, increasing the probability of generated additions or omissions.

5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

Counting hallucination is a factuality error, which motivates the following diagnostic question: *Does improvement in a perceptual metric, such as lower FID, necessarily imply a lower counting hallucination rate because the generated distribution is closer to the training distribution?* Our empirical answer is **not necessarily**.

We test this relationship by correlating CHR and NCFR with perceptual metrics. The analysis uses FID (Heusel et al., 2017) as a distribution-level metric and MUSIQ (Ke et al., 2021) as an instance-level quality metric. We estimate association with Pearson correlation (Lee Rodgers & Nicewander, 1988) for linear dependence, Spearman rank correlation (Spearman, 1904) for monotonic dependence, and Distance Correlation (Székely et al., 2007) for general nonlinear dependence. The following observations support this conclusion.

✦ **CHR-perceptual correlations depend on dataset and solver configuration rather than reflecting an intrinsic equivalence.** In the first and third rows of Table 2, Pearson correlations between CHR and FID change sign and magnitude across datasets: positive for SimObject, negative for

Table 2: Correlations between failure rates—counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR)—and perceptual metrics, including FID and MUSIQ. PTI-Hand results are produced by prompting a pretrained Stable-Diffusion-3.5-Medium model with ten hand-generation prompts to generate 5050 samples, matching the RealHand dataset size. PTI-Hand FID is computed between RealHand images and the generated samples. Here, p is the p-value for the statistical significance of correlation coefficient r or ρ . † denotes analyses that combine DDPM results with ODE-solver results across sampling conditions.

Metric	CHR vs. FID						NCFR vs. FID						CHR vs. MUSIQ						NCFR vs. MUSIQ					
	SimObject	RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	
Pearson (r)	0.87	0.01	-0.97	<0.01	0.20	0.69	0.99	<0.01	0.88	<0.01	0.93	<0.01	-0.02	0.95	-0.97	<0.01	-0.99	<0.01						
Pearson† (r)	0.49	0.14	-0.53	0.13	—	—	0.96	<0.01	—	—	0.46	0.20	—	—	-0.98	<0.01	—	—						
Spearman (ρ)	0.84	<0.01	-0.97	<0.01	-0.14	0.78	0.99	<0.01	0.92	<0.01	0.80	0.01	0.42	0.33	-0.88	<0.01	-0.46	0.29						
Spearman† (ρ)	0.75	0.01	-0.51	0.15	—	—	0.95	<0.01	—	—	0.40	0.28	—	—	-0.91	<0.01	—	—						
Distance Correlation (r_d)	0.89	—	0.97	—	0.47	—	0.99	—	0.91	—	0.93	—	0.42	—	0.97	—	0.99	—						
Distance Correlation† (r_d)	0.82	—	0.74	—	—	—	0.95	—	—	—	0.71	—	—	—	0.97	—	—	—						

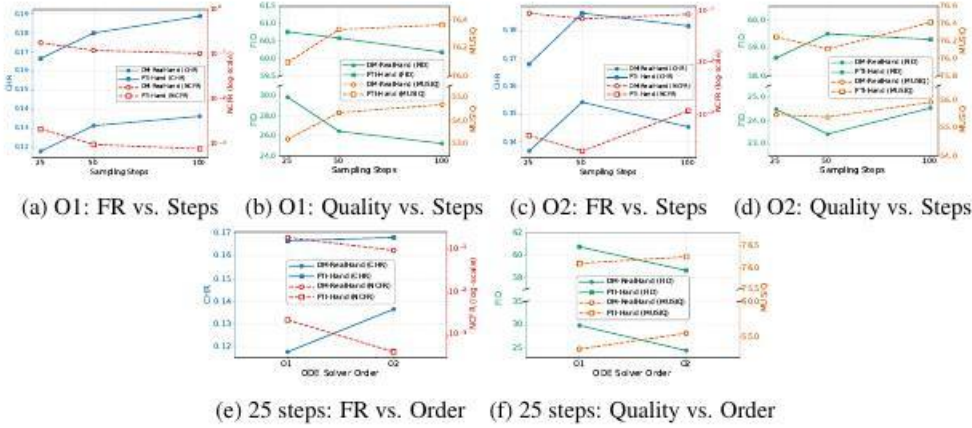


Figure 3: Trend-consistency comparison between DM-RealHand (ours) and PTI-Hand (pretrained Stable-Diffusion-3.5-Medium). O1 and O2 denote first- and second-order ODE solvers, respectively: DPM-solver-1 / DPM-solver-2 for DM-RealHand and Euler / Heun for PTI-Hand. FR metrics comprise CHR and NCFR; quality metrics comprise FID and MUSIQ. Across changes in sampling steps and solver order, the two model settings exhibit highly consistent trend patterns.

RealHand, and weakly positive for PTI-Hand. The corresponding Spearman correlations shift from positive to negative to negative. Distance correlations also decrease from 0.82 for SimObject to 0.74 for RealHand and 0.47 for PTI-Hand, indicating that the association between CHR and FID weakens as generative content becomes more complex. When DDPM results are integrated with ODE-solver results, the correlations are further attenuated.

✦ **NCFR-perceptual correlations are consistently strong and positive.** Unlike CHR, NCFR exhibits robust positive association with perceptual metrics across the evaluated settings. For NCFR versus FID, all reported correlation coefficients exceed 0.88 and have very small p-values, indicating statistically significant dependence.

5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

We next evaluate whether the RealHand findings transfer to a large-scale pretrained text-to-image model. Specifically, we assess Stable-Diffusion-3.5-Medium (Esser et al., 2024) with the counting-ready indicator and counting model defined for RealHand in Sec. 4.2.3. Additional protocol details are provided in Appendix A.8. As shown in Figure 3, the effects of solver type and sampling-step budget on counting hallucinations in the pretrained model closely match the trends observed in the controlled RealHand experiments. This consistency supports the validity of the findings in Sec. 5.2 and indicates that the evaluation protocol applies beyond models trained only on CountHalluSet. The correlation results in Table 2 provide further evidence: CHR remains weakly correlated with perceptual metrics, whereas NCFR remains strongly correlated with perceptual metrics. Together, these results indicate

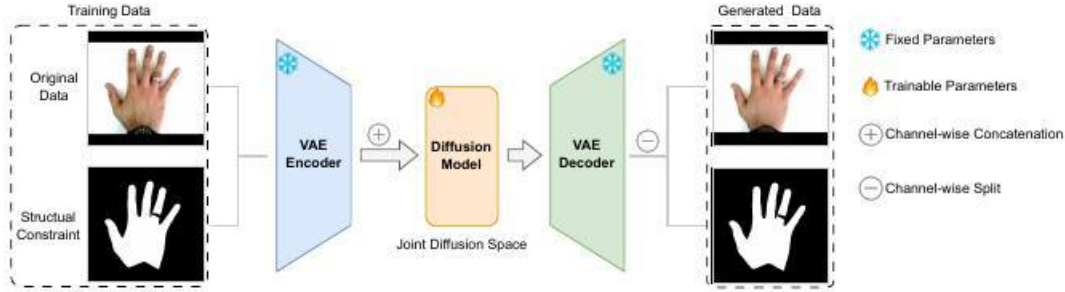


Figure 4: Schematic of the proposed joint-diffusion model (JDM). The model concatenates the latent representation of each hand RGB image with the latent representation of its corresponding segmentation mask along the channel dimension. Joint denoising of appearance and structural masks encourages anatomically plausible hand pose and finger topology, thereby reducing CHR and NCFR.

Table 3: Failure-rate comparison between the baseline LDM and the proposed JDM on RealHand across ODE solvers and sampling-step budgets. Lower values indicate better performance. Although JDM jointly generates hand RGB images and segmentation masks, FID is computed only on RGB outputs to preserve comparability with the LDM baseline.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
LDM	12.95	13.85	14.55	14.48	15.99	15.43	18.06	12.51	10.63	9.33	7.22	8.94	29.86	26.43	25.21	24.48	23.42	24.54
JDM (ours)	11.41	10.67	10.51	10.12	9.66	10.07	2.94	2.42	2.29	3.51	4.63	2.26	37.69	35.51	34.68	34.63	32.07	33.89

that the proposed protocol transfers reliably to pretrained text-to-image generation and that the main empirical conclusions are robust across model scale and conditioning regime.

5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

Observation 5 in Sec. 5.2 indicates that diffusion models preserve counting-based factuality more reliably when the countable structure has low morphological complexity, such as an object mask. This motivates a structural-conditioning question: can hallucination rates be reduced if the denoising model jointly represents RGB appearance and an explicit structural channel, such as a mask that encodes coarse topology and count information, in a shared latent space? We call models that jointly learn visual representations and structural constraints during denoising **joint-diffusion models** (JDM).

Our JDM implementation concatenates the latent representation of the hand mask channel-wise with the latent representation of the corresponding RGB image, as illustrated in Fig. 4. We then fine-tune the CelebA-HQ-pretrained UNet (Karras et al., 2018), using the RealHand configuration in Sec. 5.1, on RealHand images and their paired masks extracted by SAM-2 (Ravi et al., 2024). To represent both modalities, we expand the input and output channels of the denoising UNet to match the combined image-mask latent dimensionality (i.e., $3+3=6$ channels).

As reported in Table 3, JDM achieves lower CHR and NCFR than the vanilla LDM across the evaluated sampling conditions, indicating that joint structural denoising improves generation of anatomically intricate objects with count ambiguity. JDM also produces higher FID, which suggests a trade-off: jointly modeling RGB images and masks under a small-scale training regime increases the denoising burden and can reduce perceptual fidelity while improving structural validity. We additionally observe promising CHR reductions when applying JDM to large-scale pretrained diffusion models under carefully controlled fine-tuning settings. Further details, results, and discussion are provided in Appendix A.9.

6 CONCLUSION

This work formalizes and measures a concrete subclass of diffusion-model hallucination: violations of dataset-level object-count constraints. We introduce **CountHalluSet** and a reproducible protocol

for estimating counting hallucination rates across synthetic shapes, rendered objects, and real hand images. Controlled experiments show that standard numerical improvements in sampling, such as more ODE steps or higher solver order, do not necessarily reduce counting hallucinations for structurally complex objects such as human hands. Correlation analyses further demonstrate that perceptual metrics, including FID and MUSIQ, do not consistently encode count-specific factual errors. Experiments with pretrained text-to-image models support the broader applicability of the protocol and indicate that explicit structural denoising, as instantiated by JDM, is a promising direction for reducing counting hallucinations while preserving the underlying generative task.

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A APPENDIX

A.1 PROOF OF THE BOUND OF TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

For the joint distributions induced by the ideal reverse process $q(\mathbf{x}_{T:0})$ and the learned reverse process $p_\theta(\mathbf{x}_{T:0})$, the KL divergence is defined as

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

By the Markov property, the two reverse-process joint distributions factorize as follows:

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

Applying the KL chain rule gives the following decomposition:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[\mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[\log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

By the Data Processing Inequality (Cover & Thomas, 2012), marginalizing variables cannot increase KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

Combining Eq. (A.4) and Eq. (A.5) yields the bound on total accumulated distributional error:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

Equality can hold only in the idealized case where $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$ and $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ for every t and $\mathbf{x}_t$; this condition corresponds to exact recovery of the underlying data distribution $q(\mathbf{x}_0)$ by the diffusion model. This upper bound on the overall KL divergence ($\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0))$) between true data and model-generated data matches the DDPM variational bound (Ho et al., 2020) after excluding the data entropy term $H(\mathbf{x}_0)$. Thus, optimizing the DDPM variational objective tightens this explicit discrepancy bound between the true and generated data distributions.

Diffused-prior gap. The first error source is the mismatch between the ground-truth terminal diffusion distribution $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T|\mathbf{x}_0)$ and the sampling prior $p_\theta(\mathbf{x}_T)$. This mismatch arises because practical models use a finite number of diffusion steps T . The discrepancy can be quantified by a distributional metric such as

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

where $\mathcal{D}_{\text{KL}}$ denotes Kullback–Leibler (KL) divergence. Other distributional metrics, including FID or Wasserstein distance, may also be used to assess the same mismatch. The diffused-prior gap is present at initialization and propagates through the reverse trajectory, thereby contributing a persistent component to the final output error.

Noise model prediction error. The second error source is the approximation error introduced by the learned noise predictor. Although the function class can approximate the target noise in principle under the *Universal Approximation Theorem* (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022), the trained estimate $\epsilon_\theta(\mathbf{x}_t, t)$ generally differs from the true noise ϵ . We denote this residual by $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$. By Eq. (5), an error in ϵ_θ induces the following corresponding perturbation of the predicted mean at reverse step t :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\alpha_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Because each reverse transition depends on the mean produced at the preceding step, this bias accumulates along the denoising trajectory and contributes to the final prediction error.

A.2 PROOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

Starting from Eq. (7), we analyze a representative one-step numerical solver that approximates the ODE solution. Let $\mathbf{x}^*(\tau)$ denote the ideal trajectory that satisfies this ODE with the true score function $h^*(\mathbf{x}(\tau), \tau)$ and starts from a ground-truth initial condition $\mathbf{x}^*(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}_T)$. Similarly, $\hat{\mathbf{x}}_k$ denotes the numerical trajectory computed by an ODE solver at discrete timesteps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, and $\hat{h}$ denotes the estimated model function. For a given numerical trajectory from $\hat{\mathbf{x}}_0$ to $\hat{\mathbf{x}}_N$, the objective is to express its total accumulated trajectory error. Let $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$ be the trajectory error at step k . Consider the transition from τ_k to $\tau_{k+1} = \tau_k - \Delta t$.

Ideal path evolution. The exact one-step evolution from τ_k to τ_{k+1} under the variation-of-constants formula is

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear part $\frac{dz}{dt} = f(\tau)z$. The integration limits reflect the reverse-time orientation of the sampling trajectory.

Numerical path evolution. A one-step numerical solver approximates this evolution, and its update differs from the exact trajectory generated by $\hat{h}$ when initialized at $\hat{\mathbf{x}}_k$. Let $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h})$ be the exact solution at τ_{k+1} of the ODE using $\hat{h}$, starting from $\hat{\mathbf{x}}_k$ at τ_k :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The numerical method introduces a local truncation error δ_k :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} | \hat{\mathbf{x}}_k, \hat{h}) + \delta_k. \quad (\text{A.11})$$

Consequently, the numerical update can be written as

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

Error recurrence. Subtracting the ideal evolution in Eq. (A.9) from the numerical evolution in Eq. (A.12) gives

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

Accumulated error formulation. The recurrence can be unrolled from $k = 0$ to $N - 1$. Let $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}), \dots, \mathcal{G}(\tau_{k+1}, \tau_k)$ for $j > k$, and $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$. Repeated application of the recurrence yields

$$\begin{aligned} e_N &= \mathcal{G}(\tau_N, \tau_0) e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \\ &\quad \cdot \left(\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \end{aligned} \quad (\text{A.14})$$

Substituting $\tau_N = 0$, $\tau_0 = T$, $e_N = \hat{\mathbf{x}}_N - \mathbf{x}^*(0)$, and $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$:

$$\hat{\mathbf{x}}_N - \mathbf{x}^*(0) = \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \cdot \left(\underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \quad (\text{A.15})$$

where the propagated initial-error term is the deterministic contribution of a particular initialization error to the final path error; statistically, this term originates from the distributional diffused-prior gap.

A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

As in Eq. (2), the DDPM (Ho et al., 2020) forward process can be written as

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

The corresponding reverse update is

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \frac{1 - \alpha_t}{\sqrt{\alpha_t} \sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon. \quad (\text{A.17})$$

For DDIM (Song et al., 2021), the forward process is identical to Eq. (A.16); its deterministic reverse process with $\eta = 0$ is

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon_\theta(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \left(\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \epsilon_\theta(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Although DDPM and DDIM share the same forward diffusion process, DDIM is not obtained merely by removing the stochastic term from the DDPM reverse process. The two samplers use distinct coefficient parameterizations for the predicted-noise term.

A.4 DATASET EXAMPLES AND DETAILS

Figure A1 shows representative samples from the three datasets used in the study. Table A1 reports category-level counts and indicates that each dataset is approximately balanced across its evaluated categories. Table A2 reports image counts by object multiplicity and shows comparable coverage across count configurations, reducing object-count bias. The datasets use the same ordering in both tables, and the category order in Table A1 is

- **ToyShape**: triangle, square, pentagon;
- **SimObject**: mug, apple, clock;
- **RealHand**: finger.

A.5 EXAMPLES OF COUNTING HALLUCINATIONS

Fig. A2 provides representative counting hallucinations for all three datasets. In ToyShape and SimObject, hallucinated samples either contain more than one instance of at least one category or contain no foreground object despite an otherwise normal background, as shown in Fig. A2(a) and Fig. A2(b). In RealHand, hallucinated samples are defined by invalid finger counts, such as 4 or 6 fingers, as shown in Fig. A2(c). The final two images in the first row of Fig. A2(c) visually resemble five-finger cases, but one finger is incomplete and artifacted, yielding only four valid fingers; these samples are therefore labeled as counting hallucinations.

Table A1: Category-wise object counts for each dataset.

	ToyShape	SimObject	RealHand
Category 1	20,027	19,916	25,250
Category 2	19,902	20,101	-
Category 3	19,978	19,974	-

Table A2: Image counts grouped by object multiplicity for each dataset.

	ToyShape	SimObject	RealHand
1 object	10,003	10,022	-
2 objects	10,087	9,965	-
3 objects	9,910	10,013	-
5 objects	-	-	25,250
Total	30,000	30,000	25,250

A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

Figure A3 presents representative YOLO predictions on generated RealHand images. The detector identifies fingertips across diverse hand poses and orientations, including dorsal and palmar views. However, purely count-based detection can fail to distinguish anatomically implausible configurations from anatomically valid ones when the fingertip count remains five. Examples include duplicated thumbs (e.g., the penultimate sub-figure of the first row and the last sub-figure of the last row in Fig. A3(a)) or hands with five middle fingers but no thumb (e.g., the last sub-figure of the first row and the penultimate sub-figure of the last row in Fig. A3(a)). These anatomically abnormal but count-valid samples are still detected as five-finger hands and therefore remain indistinguishable under a finger-count-only criterion.

Figure A3(b) shows counting-hallucinated samples for which YOLO predicts invalid fingertip counts, including 3, 4, 6, or 7 fingertips. Figure A3(c) illustrates rare edge cases (<1%) in which unclear boundaries, cracked fingers, or floating artifacts cause detection failures. These cases can be partially mitigated by changing YOLO inference hyperparameters, such as IoU and confidence thresholds, but such changes may reduce accuracy on the majority of counting-valid samples. Because the same detector configuration is used for every RealHand evaluation, these rare failures do not change the comparative conclusions, although fully resolving them remains technically difficult and empirically low-yield.

A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

Figure A4 shows representative generated hand images from the three evaluation classes: counting-valid, counting-hallucinated, and non-counting failure. As illustrated in Fig. 2, the RealHand protocol first applies the counting-ready indicator to separate countable images, comprising counting-valid and counting-hallucinated samples, from images unsuitable for counting. Non-counting failures typically contain severe artifacts such as distorted, cracked, partially visible, or floating fingers. Countable samples are then classified as valid or hallucinated according to whether the detected number of valid fingertips equals five.

A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

In the PTI-Hand experiments, we use 10 hand-generation prompts for Stable-Diffusion-3.5-Medium. The prompts are listed below.

1. A single human hand, front view, all fingers in frame and visible.
2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.

4. Overhead shot of a person’s hand, entire hand in frame, all fingers visible, plain background.
5. One hand, top view, five fingers visible, simple background.
6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

The hallucination definition assumes that the model produces configurations absent from the training distribution. For large pretrained diffusion models, this condition cannot be verified directly because the full training distribution is unavailable. We therefore use a conservative operational criterion: any generated hand image with six or more fingers is labeled as counting-hallucinated. This criterion is appropriate because real images may contain fewer than five visible fingers due to occlusion, articulation, or viewpoint, whereas hands with six or more fingers are highly unlikely in natural image data. Thus, finger counts ≥ 6 constitute a reliable and empirically verifiable signal of counting hallucination. Generated examples are shown in Fig. A5.

A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

For the JDM experiments in Table 3, the denoising UNet is fine-tuned for 120k steps, whereas the vanilla LDM is fine-tuned for 80k steps. The larger optimization budget is required because expanding the latent representation from three to six channels introduces a substantial mismatch relative to the pretrained LDM latent distribution. Additional optimization is therefore needed to reduce this distributional mismatch and make the JDM capacity effective.

A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

Table A3: Failure-rate and image-quality comparison between fine-tuned SD-1.5 and fine-tuned SD-1.5 with the joint-diffusion training paradigm on RealHand. The baseline row reports Stable-Diffusion-1.5 fine-tuned on RealHand. The second row reports the configuration that freezes all intermediate layers of the baseline model and fine-tunes only the input and output layers after extending the channel dimensions. Lower values indicate better performance.

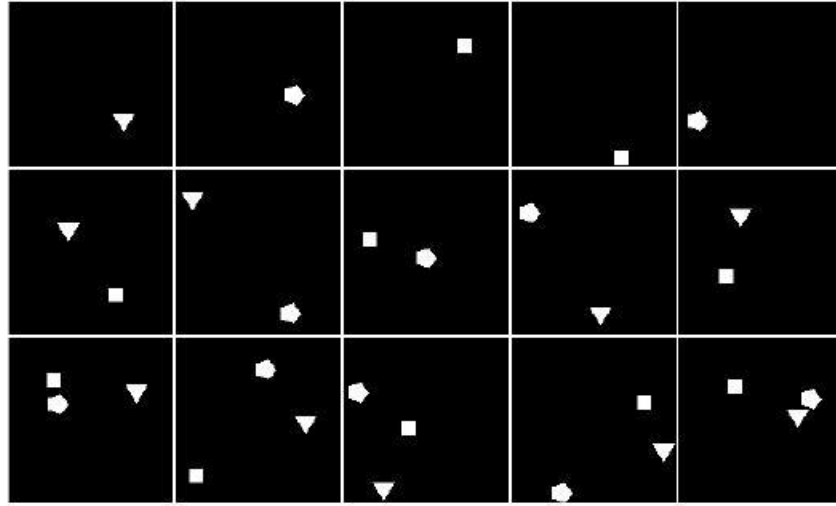
Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
SD-1.5	0.3362	0.3404	0.3630	0.3396	0.3376	0.3362	0.0039	0.0032	0.0020	0.0059	0.0026	0.0018	61.51	57.71	57.70	58.53	57.57	56.69
SD-1.5 with joint-diffusion fine-tuning	0.2838	0.3222	0.4145	0.3053	0.3277	0.3558	0.0947	0.0511	0.0487	0.0549	0.0362	0.0327	70.65	79.22	102.39	70.80	72.84	83.07

JDM should be interpreted as a **training paradigm** for improving factual generation in diffusion models rather than as a distinct inference algorithm. Because JDM only expands the input and output channels of the UNet while preserving the intermediate layer widths, it introduces **negligible inference-time overhead relative to vanilla LDM**. However, adding mask channels to a pretrained text-to-image model **changes the manifold geometry of the pretrained latent space and perturbs the original denoising distribution**, which makes direct fine-tuning technically nontrivial.

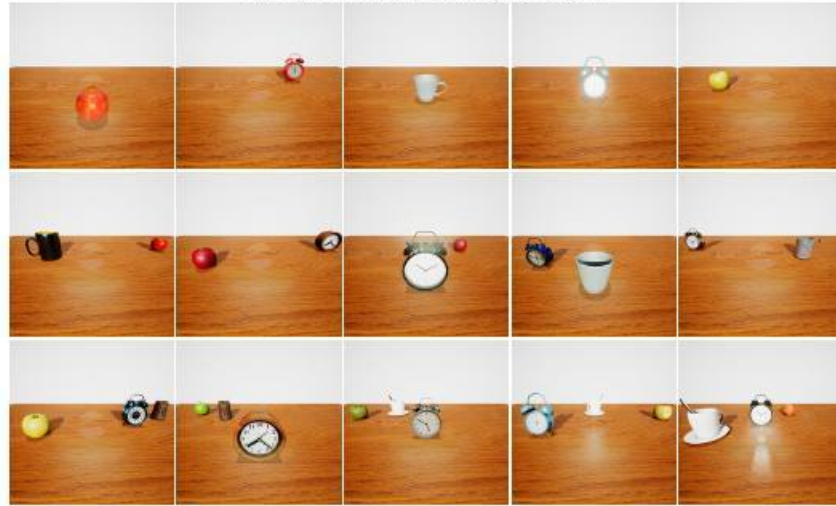
Despite this difficulty, JDM still reduces counting hallucinations for large pretrained diffusion models under controlled fine-tuning conditions. In practice, aligning the extended JDM latent space with the rigid latent geometry of a large pretrained model is challenging at the scale of RealHand. We therefore use a two-stage fine-tuning strategy: first fine-tune Stable-Diffusion-1.5 on RealHand, then freeze all intermediate UNet layers and fine-tune only the input and output layers with extended

channels. As shown in Table A3, even this constrained setup yields consistently lower counting hallucination rates at 25 and 50 sampling steps for both solvers.

Overall, the evidence indicates that JDM is a promising mechanism for improving factual consistency in image generation. **Evaluating its behavior under genuinely large-scale training remains an important direction for future work.**



(a) Representative ToyShape samples

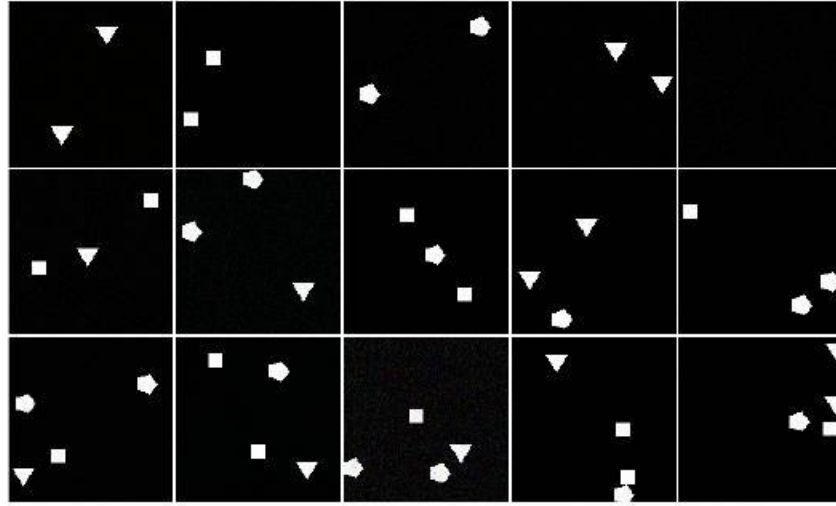


(b) Representative SimObject samples

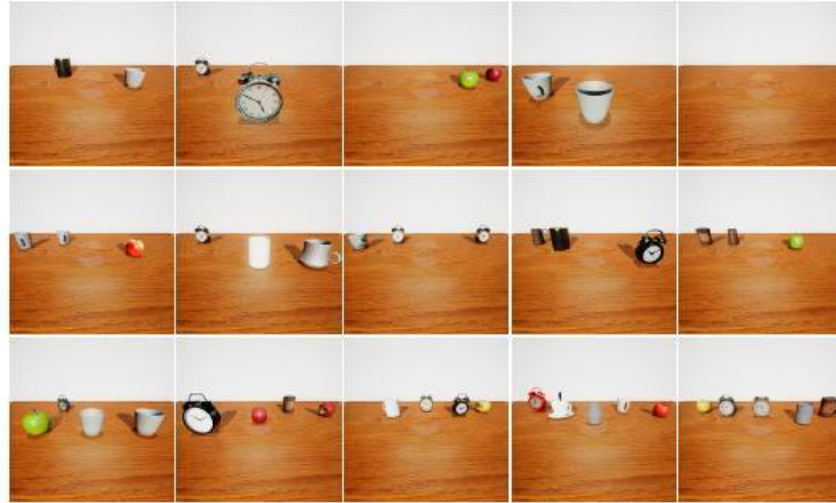


(c) Representative RealHand samples

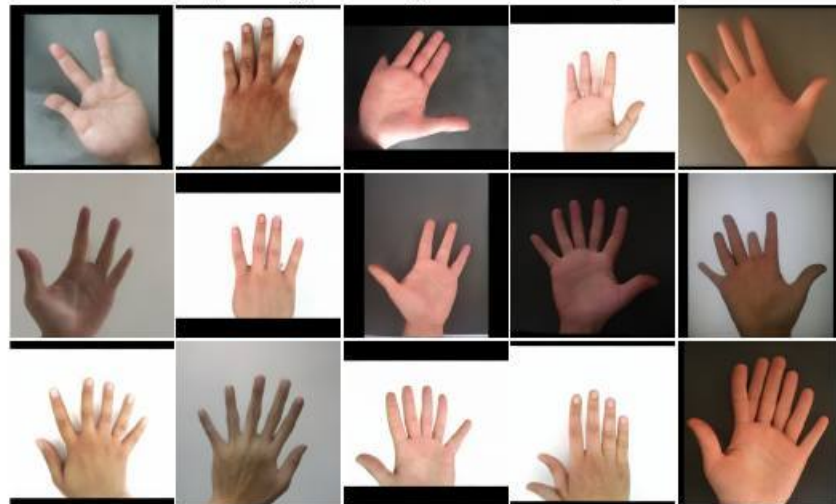
Figure A1: Representative samples from the CountHalluSet suite: ToyShape, SimObject, and RealHand.



(a) ToyShape counting-hallucination examples

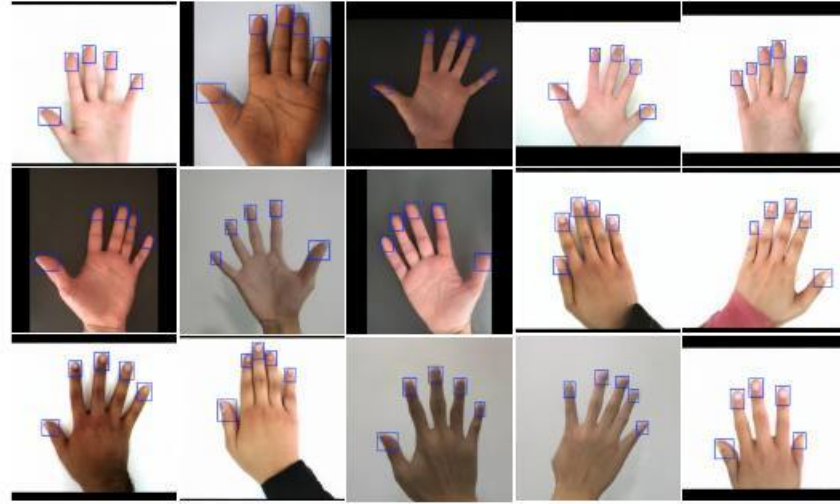


(b) SimObject counting-hallucination examples

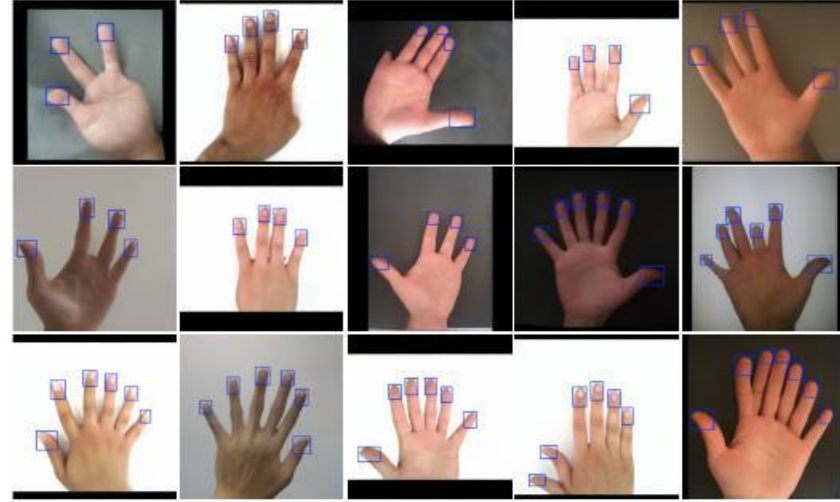


(c) RealHand counting-hallucination examples

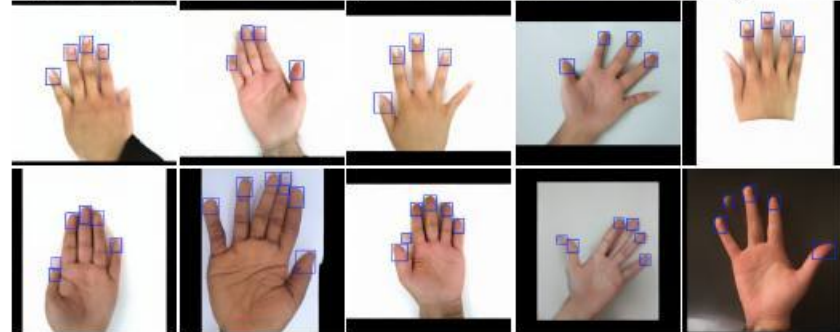
Figure A2: Representative counting hallucinations for ToyShape, SimObject, and RealHand.



(a) YOLO predictions for count-valid five-finger samples



(b) YOLO predictions for counting-hallucinated samples with non-five-finger counts

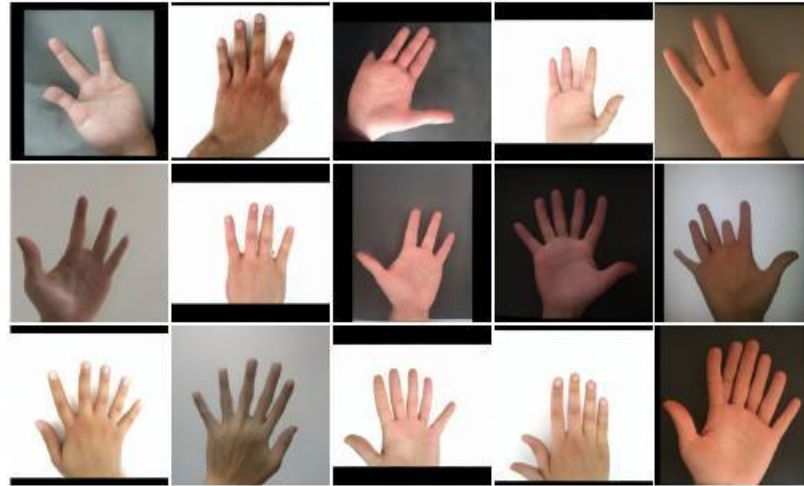


(c) Rare edge cases with YOLO prediction failures

Figure A3: YOLO predictions on generated RealHand images. (a) Counting-valid samples for which YOLO predicts five fingertip boxes, including cases with two thumbs. (b) Counting-hallucinated samples with invalid fingertip counts, such as 3, 4, 6, or 7 detected fingertips. (c) Rare edge cases not filtered by the counting-ready indicator in which YOLO struggles with blurred, incomplete, or anatomically disconnected fingers.



(a) Counting-valid samples with five valid fingers



(b) Counting-hallucinated samples with non-five-finger counts



(c) Non-counting failure samples unsuitable for counting

Figure A4: Three evaluation classes for generated RealHand images. (a) Counting-valid images contain five clearly discernible valid fingers. (b) Counting-hallucinated images contain an invalid number of valid fingers, such as 4 or 6. (c) Non-counting failures are unsuitable for counting because of severe visual artifacts, including distorted, cracked, partially visible, or floating fingers.



(a) Counting-valid samples with five valid fingers



(b) Counting-hallucinated samples with more than five valid fingers



(c) Non-counting failure samples unsuitable for counting

Figure A5: Three evaluation classes for hand images generated by pretrained Stable-Diffusion-3.5-Medium (PTI-Hand). (a) Counting-valid images contain five clearly discernible fingers. (b) Counting-hallucinated images contain more than five valid fingers, such as 6 or 7. (c) Non-counting failures are unsuitable for counting because of severe visual artifacts, including distorted, cracked, partially visible, or floating fingers.